Transmission medium: A transmission medium can be defined as anything that can carry information from a source to a destination. The transmission medium is usually free space, metallic cable or fiber optic. Communication is possible only if the information is encoded into a signal.

Factor affecting Data communication of a medium:

1. Communication Bandwidth
2. The transmission Impairments

Bandwidth: The available path of communication is called Bandwidth. The difference between highest and lowest frequencies from a frequencies band is referred to as Bandwidth.

Transmission Impairments:

1. Attenuation
2. Distortion
3. Noise

Attenuation: Attenuation means loss of energy. When a signal travels through a medium, it loses some of its energy in overcoming the resistance of the medium.

Distortion: Distortion means that the signal changes its form or shape. Distortion can occur in a composite signal.

Noise: Noise is the major causes of impairments. Several types of noise like thermal noise, induced noise, crosstalk and impulse noise may corrupt the signal.

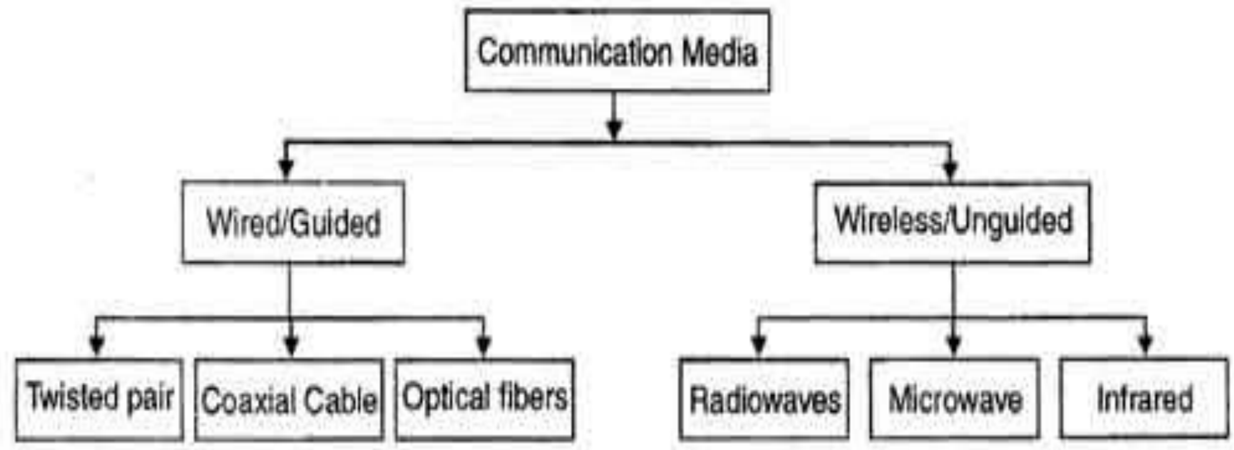
Thermal noise is the random motion of electron which creates an extra signal not originally sends by the sender.

Induced noise comes from different source like motors, sending antenna, receiving antenna, transmission medium.

Crosstalk is the effect of one wire on the other.

Impulse noise comes from power line, lighting source.

Types of transmission media:



Guided Media:

Guided Transmission Media uses a "cabling" system that guides the data signals along a specific path. Guided media are those that provide a conduit from one device to another using different cable.

There four basic types of Guided Media:

1. Twisted Pair
2. Coaxial Cable
3. Optical Fiber

Twisted paired cable characteristics:

- Requires amplifiers for analog signals.
- Requires repeaters for digital signals.
- Attenuation is a strong function of frequency.
- Higher frequency implies higher attenuation.
- Twisting reduces low frequency interference.
- Different twist length in adjacent pairs reduces crosstalk.

The application of Twisted paired cable:

- Most common transmission media for both digital and analog signals.
- TP cables are used in telephone lines to provide voice and data channels.
- The line that connects subscribers to the central telephone office is most commonly UTP cable.
- The DSL lines that are used by the telephone companies to provide high data rate connections also use high bandwidth capability UTP cable.
- Local Area Network (LAN) also uses twisted-pair cable.

Coaxial Cable characteristics:

- Used to transmit both analog and digital signals.
- Superior frequency characteristics compared to twisted pair.
- Can support higher frequencies and data rates.
- Less susceptible to interference and crosstalk than twisted pair.
- Constraints on performance are attenuation, thermal noise, and intermodulation noise.
- Requires repeaters every few kilometers for long distance transmission.

The application of Coaxial cable:

- Analog telephone networks.
- Most common use is in cable TV.
- 10 Base-2, or Thin Ethernet, uses RG-58 coaxial cable with BNC connectors to transmit data at 10 Mbps with a range of 185 m.
- 10Base5, or Thick Ethernet, uses RG-11 to transmit 10 Mbps with a range of 5000 m.

Fiber optic cable characteristics:

- The diameter of glass core in single mode fiber is very small ranging from 8 to 10 microns.
- In this mode, light can propagate only in a straight line, without bouncing.
- Fiber glass has lower density (index of refraction) that creates a critical angle close enough to 90° such that the beam propagates in a straight line.
- In this case, propagation of different beams is almost identical and delays are negligible. The beams arrive at destination together and can be recombined with little distortion to the signal.
- Single mode fibers are more expensive and are widely used for long distance communication.
- These types of fibers can transmit data at 50 Gbps for 100 kilometers without amplification.

Applications of fiber-optic cable

- Fiber-optic cable is often found in backbone networks because its wide b/w is cost-effective. SONET n/w provides such backbone.
- Some cable TV companies use a combination of optical-fiber and coaxial cable.
- Telephone companies also using optical-fiber cable.
- Local Area Networks (LANs) such as 100BaseFx network (Fast Ethernet) and 1000Base-X also use fiber-optic cable.

Advantages of fiber-optic cable

1. Higher Bandwidth: Higher data rate than TP & coaxial cable.

2. Less signal attenuation: Fiber-optic transmission distance is significantly greater than that of other guided media. A signal can run for 50 km without requiring regeneration. We need repeaters after every 5km for coaxial or TP cable.

3. Noise resistance: Because fiber-optic transmission uses light rather than electricity, noise is not a factor. External light, the only possible interference, is blocked from the channel by the outer jacket.

4. Light weight: Fiber-optic cables are much lighter than copper cables.

5. More immune to tapping (or Security): Fiber-optic cables are more immune to tapping than copper cables. Copper cables create antennas that can easily be tapped.

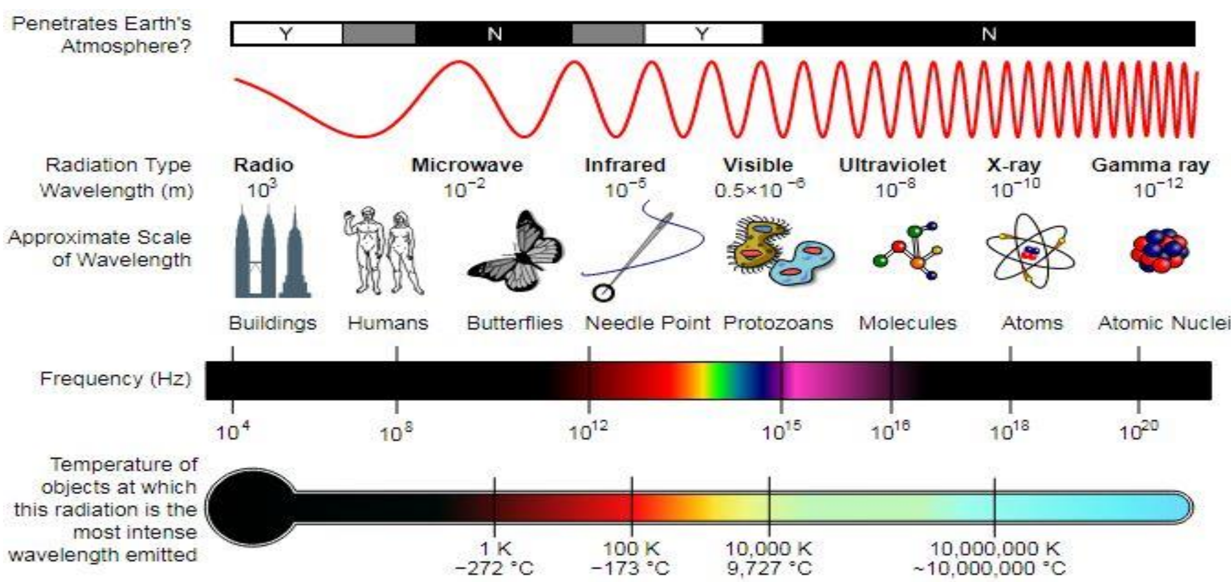
6. Optical fiber can carry thousands of times more information than copper wire. For example, a single-strand fiber strand could carry all the telephone conversations in the United States at peak hour. Fiber is more lightweight than copper. Copper cable equals approximately 80 lbs/1000 feet while fiber weighs about 9 lbs/1000 feet.

7. Reliability: Fiber is more reliable than copper and has a longer life span.

8. Fiber optic cable can carry signals for longer distance without repeater than co-axial cable.

Disadvantages of fiber-optic cable.

1. **Installation/maintenance expertise:** Installation and maintenance need expertise that is not yet available everywhere.
2. **Unidirectional:** Propagation of light is unidirectional.
3. **Cost:** Fiber-optic cable is more expensive.
4. **Fragility:** Glass fiber is more easily broken than wire, making it less useful for applications where h/w portability is required.
5. **Limited physical arc of cable of cable.** Bend it too much and it will break!



Radio Wave:

1. Low energy wave with long wavelength
2. Lowest frequency among all the electromagnetic wave.
3. Includes AM, FM, radar, TV waves.
4. The range of frequencies between 3 KHz to 1GHz.
5. Used omnidirectional antenna
6. Its propagation is sky propagation mode.
7. It can penetrate wall.

Microwave:

1. Wavelength is shorter than radio wave.
2. Frequency is longer than radio wave.
3. Used unidirectional antenna.
4. The range of frequency between 1 GHz to 300 GHz.
5. It propagates in line to sight propagation.
6. VHF microwave can’t penetrate wall.
7. Satellite network, cellular phone, micro oven.

Infrared:

1. Infrared has lowest wavelength.
2. Higher frequency than microwave.
3. The frequency range 300 GHz to 400 THz.
4. It can’t penetrate wall.
5. It is useless in long distance communication.
6. It can’t work properly in sun’s ray.
7. Remote control.

Propagation:

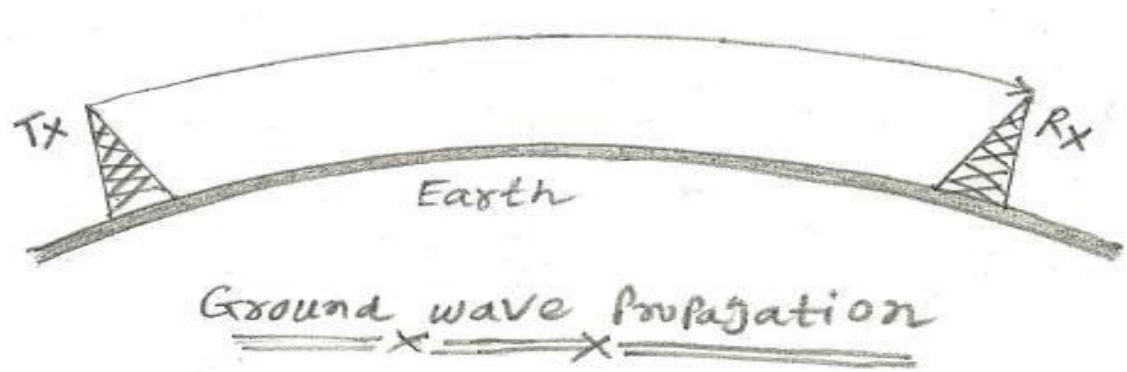
Propagation is a way in which waves travel with respect to the direction of oscillations.

Types of propagation:

1. Ground Propagation
2. Sky Propagation
3. Line of sight or Space Propagation

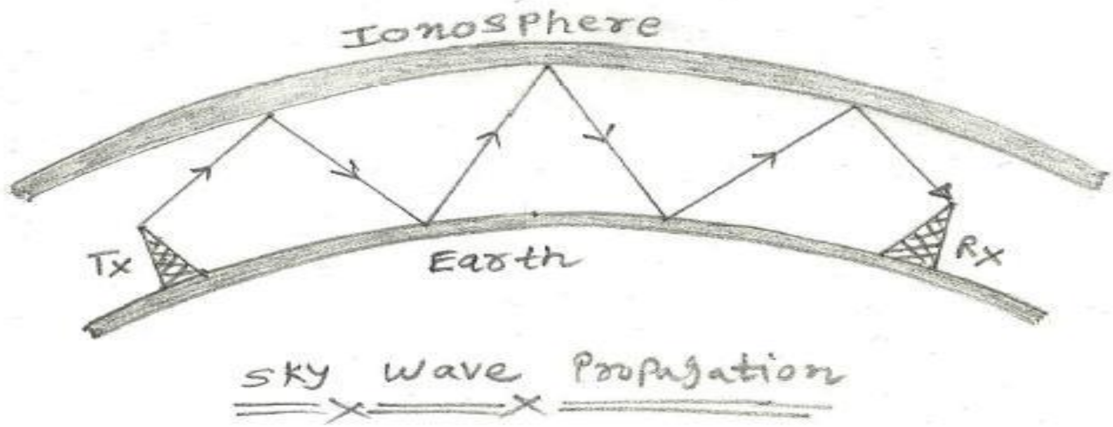
Ground Propagation:

Ground Wave propagation is a method of radio wave propagation that uses the area between the surface of the earth and the ionosphere for transmission. The ground wave can propagate a considerable distance over the earth's surface particularly in the low frequency and medium frequency portion of the radio spectrum. Ground wave radio propagation is used to provide relatively local radio communications coverage.



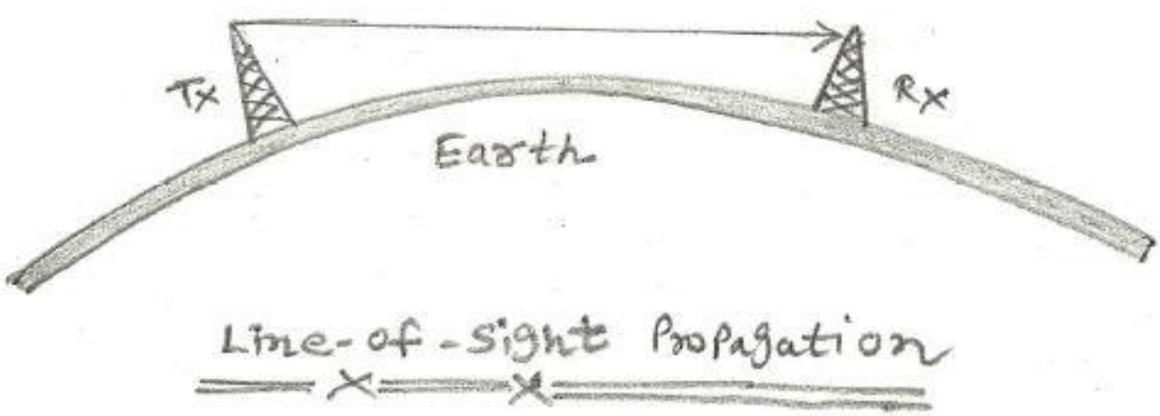
Sky Propagation:

Sky propagation refers to the propagation of radio waves that reflected or refracted back toward Earth from the ionosphere. It is mostly used in the shortwave frequency bands.



Space or Line of sight:

Line-of-sight propagation is a types of wave propagation that travel in a direct path from the source to the receiver. Electromagnetic transmission includes light emissions traveling in a straight line.



Type	Frequency	Application
Ground	VLF, LF, MF	1. Submarine Communication 2. AM, FM, Television Broadcast
Space	HF, VHF, UHF	1. Satellite Communication 2. Mobile Communication
Sky	2MHz – 30MHz	1. Microwave Link 2. Rader Communication